Quiz#2

Name(print): _____ **Student ID:** _____

1. The raven is sent from the Castle Black, which is at the origin, observing the white walkers at the coordinates $\vec{r}_4 = 3\hat{\imath} + 6\hat{\jmath}$ at the beginning of season 4, where length units are in kilofurlongs. Which of the following statements is correct?
   A. The distance between the white walkers and the Castle Black is 9 kilofurlongs.
   B. The distance between the white walkers and the Castle Black is 6 kilofurlongs.
   C. The distance between the white walkers and the Castle Black is 3 kiolofurlongs.
   D. None of the above is correct.

2. If the displacement of an object is given by the function $\vec{s} = ((1.3m/s^3)t^3 + (2.7m/s^2)t^2)\hat{\imath}$, which of the following statement is correct?
   A. This is a constant acceleration motion.
   B. The magnitude of the acceleration is 2.7 m/s^2
   C. The magnitude of the acceleration at t = 0 is 0 m/s^2
   D. The magnitude of the acceleration at t = 0s is 5.4 m/s^2

3. An object takes 3 second to move from position A ($\vec{r}_A = -4.0\hat{\imath} + 10\hat{\jmath}$) to position B ($\vec{r}_B = 2.0\hat{\imath} + 5.0\hat{\jmath}$), where length units are in meters. Which of the following statements is correct?
   A. The moving distance is 7.8 m.
   B. The speed at t = 0 must be 2.6 m/s.
   C. The acceleration at t = 0 must be 0.87 m/s^2
   D. None of the above is correct.